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To the Editors of *Applied Energy*,

We submit for your consideration the manuscript entitled:

“Seven Hundred Simulations Suffice: Matching a 900,000-Building Foundation Model through Operational Diversity in Zero-Shot Load Forecasting”

by Jeong-Uk Kim (Sangmyung University, South Korea).

Why *Applied Energy*?

This journal’s focus on energy efficiency, intelligent building systems, and data-driven methods makes it the natural home for this work. *Applied Energy* has published the foundational studies on building energy prediction, demand response, and simulation-based data generation that our paper directly builds upon and extends.

Summary of contributions

Zero-shot building load forecasting—predicting energy use for previously unseen buildings without retraining—underpins grid-scale demand response and energy management. The prevailing approach, exemplified by BuildingsBench (NeurIPS 2023), achieves state-of-the-art accuracy (13.28% NRMSE on 955 commercial buildings) by training on 900,000 building simulations.

We demonstrate that **data quality can substitute for data quantity**: 700 EnergyPlus simulations, generated by sampling 14 building archetypes with 12-dimensional Latin Hypercube Sampling (LHS) to maximise operational diversity, achieve 13.11 ± 0.17% NRMSE (five-seed mean) on the identical 955-building evaluation set—matching the 900K-building SOTA within the inter-seed standard deviation. The best seed reaches 12.93%, surpassing SOTA by 0.34 percentage points.

Four controlled findings are reported:

1. **Data design dominates scale.** A matched 700-building comparison (Korean LHS vs. BuildingsBench subsampled) shows a 1.15 pp advantage with augmentation and 1.61 pp without, attributing the gap to LHS schedule design rather than normalisation or augmentation alone.

2. **RevIN is scale-dependent.** Reversible Instance Normalisation reduces NRMSE by 1.61 pp on our 700-building corpus but increases it by 0.62 pp on the full 900K corpus—a previously unreported asymmetry with practical implications for foundation model design.

3. **Rapid N-scaling saturation.** Seventy buildings (five per archetype) already match SOTA (13.28 ± 0.12%); diminishing returns set in beyond 140 buildings, suggesting that archetype coverage matters more than sample count.
4. **Sim-to-real transfer.** Zero-shot inference on 218 real Korean convenience stores yields 12.30% versus 13.14% for BuildingsBench, providing preliminary evidence of simulation-to-reality generalisation.

Together, these findings substantially lower the data barrier for zero-shot building load forecasting and offer actionable guidance on simulation budget allocation.

Originality and ethics

This manuscript has not been published previously, is not under review elsewhere, and all named authors have approved the submission. The convenience store dataset was collected under a data-sharing agreement with the data provider (written confirmation pending final submission). Code and model weights will be made available upon acceptance via a public GitHub repository.

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We suggest the following as potential reviewers, given their expertise in building energy forecasting and foundation models:

- Researchers active in the BuildingsBench benchmark community
- Researchers specialising in data-centric AI for energy systems
- Researchers with expertise in zero-shot and transfer learning for time-series forecasting

We have no conflicts of interest to declare.

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